

The Coin Sorter Design Project

Group 1 – Internal Revenue Service

Nouh Shaikh, Habeeb Salaudeen, Pakavon So-by

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Portable Coin Sorter

This project aims to develop a compact, affordable, and portable coin sorter designed to help individuals and small businesses efficiently organize and manage loose change. Unlike existing commercial coin sorters that are often bulky, expensive, or deduct a fee, this solution will provide a low-cost, user-friendly alternative that can be used at home, in vehicles, or in workplaces such as restaurants or donation centers. Through interviews with peers and potential users, a strong need was identified for a more accessible device that accurately sorts coins by denomination based on size. Using 3D-printed components and low-cost materials, the design is expected to stay within a \$50 budget while maintaining accuracy and mobility. With a high likelihood of completion by April 2025, this project seeks to bridge the gap between large-scale financial equipment and everyday user convenience by delivering a scalable, functional prototype tailored to real-world needs.

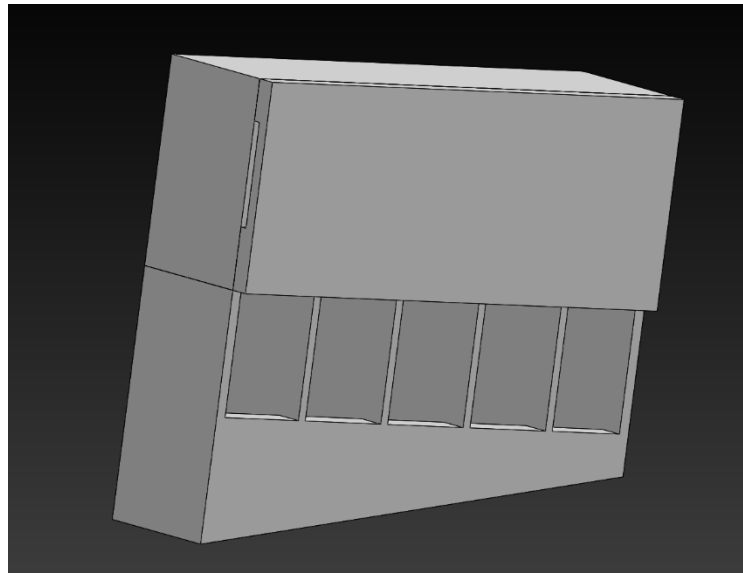


Figure 1 – Assembly

Figure 1 shows the final assembly drawing, which displays all three connected components along with their respective technical drawings. The upper section functions as a divider, directing coins into the third component, which serves as the storage unit where users can easily access the sorted coins by hand. The most critical element of the assembly is the main plate, which performs the actual sorting based on the coins' dimensions. As the coins roll through the system, they are guided by the main plate, pushed into the divider, and finally deposited into the appropriate storage compartments.

Prototype

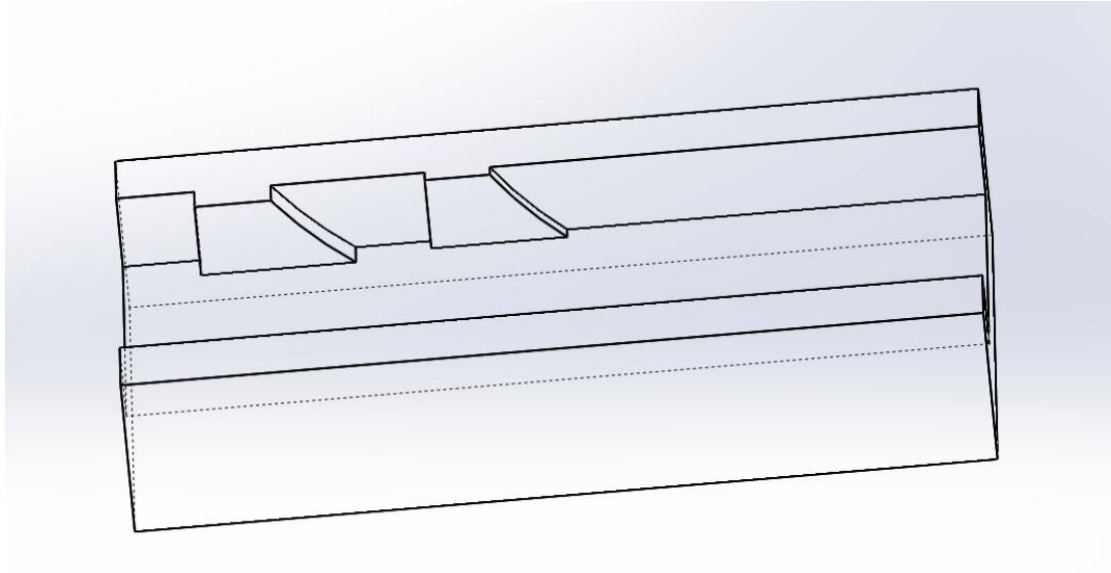
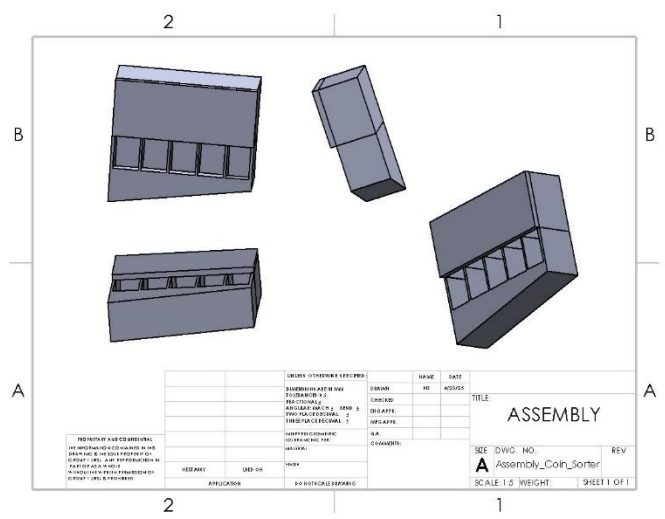
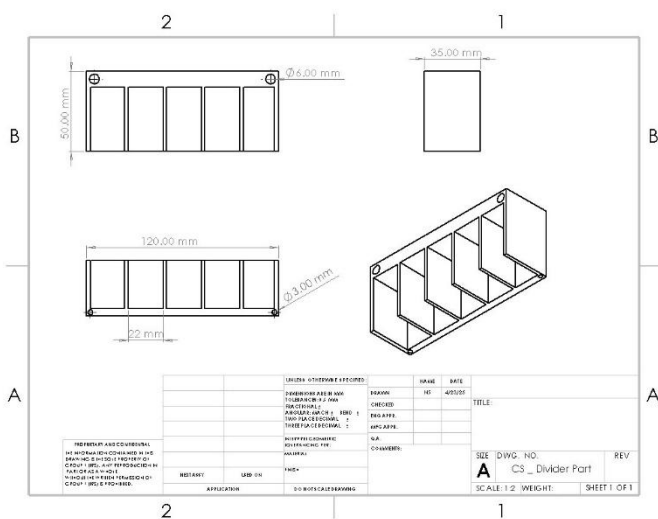
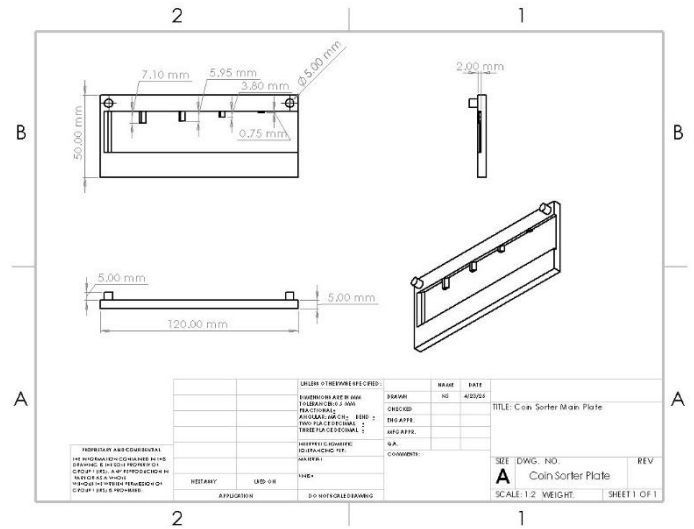
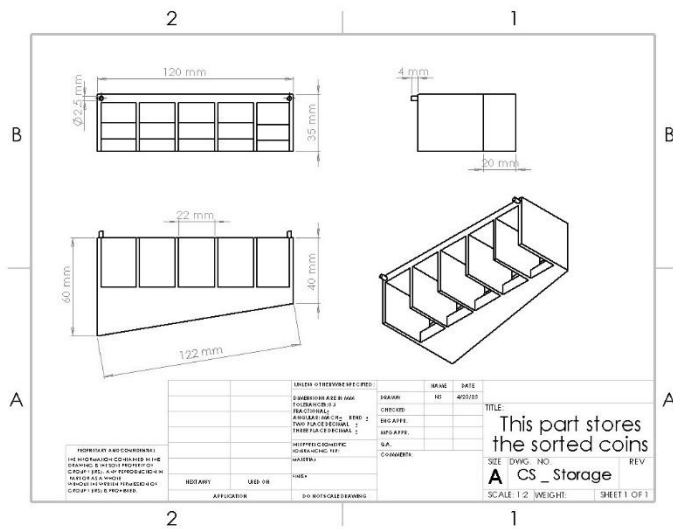


Figure 2 – Prototype (Section View)

Our current prototype utilizes the natural rolling motion of coins to achieve efficient and reliable sorting. Coins are directed along a sloped surface embedded with precisely spaced indents each calibrated to within ± 0.05 of the coin diameters to guide them out of the main chamber and into their respective denomination windows. This design was developed in response to the limitations observed in earlier versions, which were prone to jamming and mechanical failures due to more complex, drag-based mechanisms. By leveraging the inherent geometry of circular coins, this rolling-based approach reduces friction, minimizes error points, and improves overall performance. For a visual representation of this concept, please refer to the section view of the assembly within *Figure 2*.

Coin Sorter Part Drawings



Responsibilities Table

Member name(s)	Responsibilities
Habeeb Salaudeen	Management, quality assurance and production of the final product
Nouh Shaikh	Designing and updating the product using Solidworks.
Pakavon So-By	Measuring, calculating dimensions and testing products for issues.

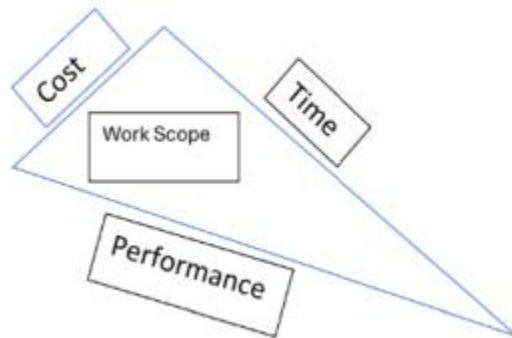
The management areas of this project will be put in the hands of the group. Habeeb will handle the management updates, assurance of product quality, and the production of the final project. All the CAD (particularly SolidWorks) design concepts will be Nouh's responsibility. Pakavon's responsibilities are the measurements of the design and testing of the products in search of issues.

Gantt Chart



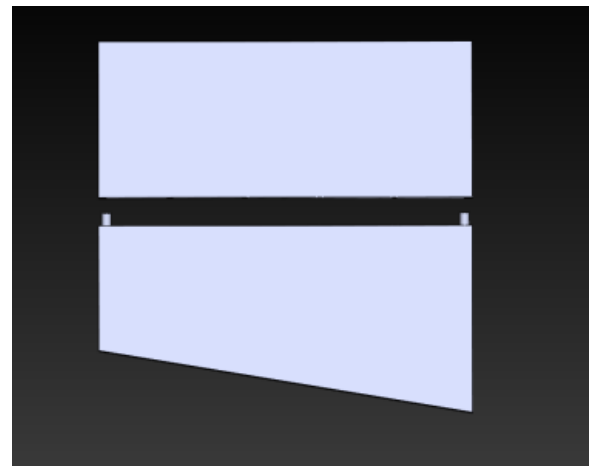
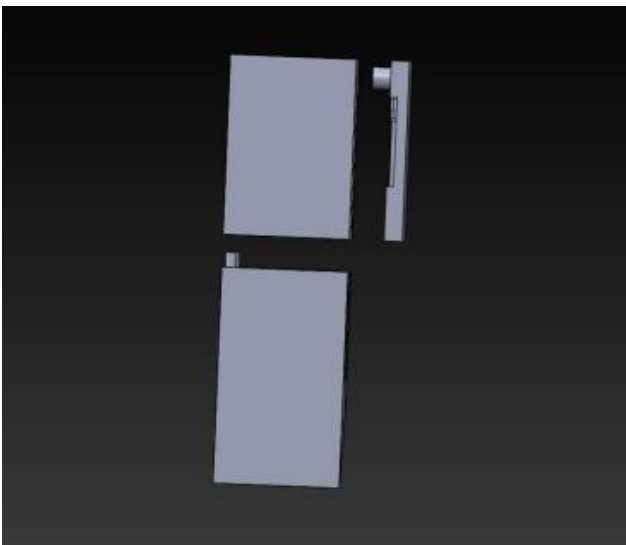
This Gantt chart provides dates when the group will manage our respective responsibilities. Nouh is expected to manage the CAD for most of the time since the product design derives from it. Habeeb is expected to do quality maintenance most of the time, with final 4 production after that. Pakavon will handle dimension calculations and, shortly after, will focus on testing the product.

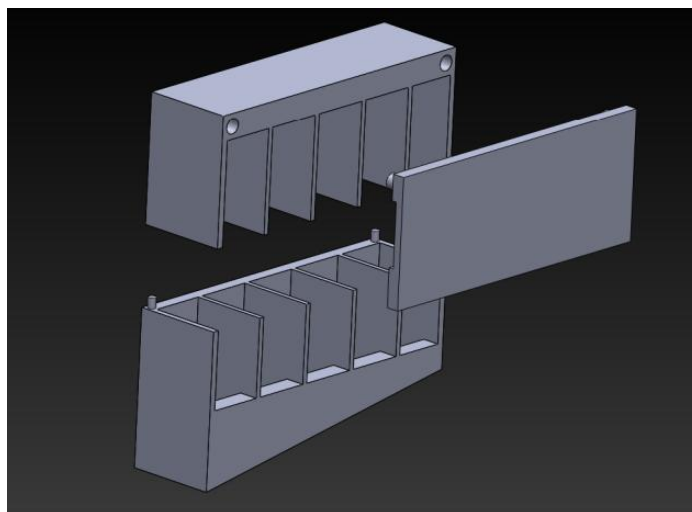
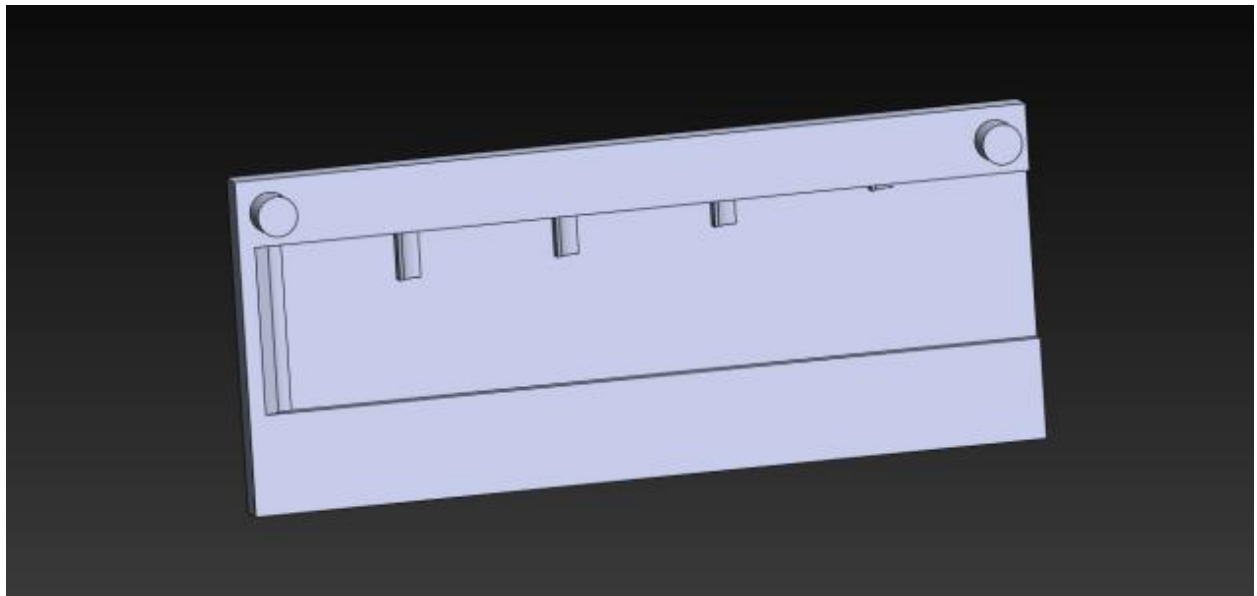
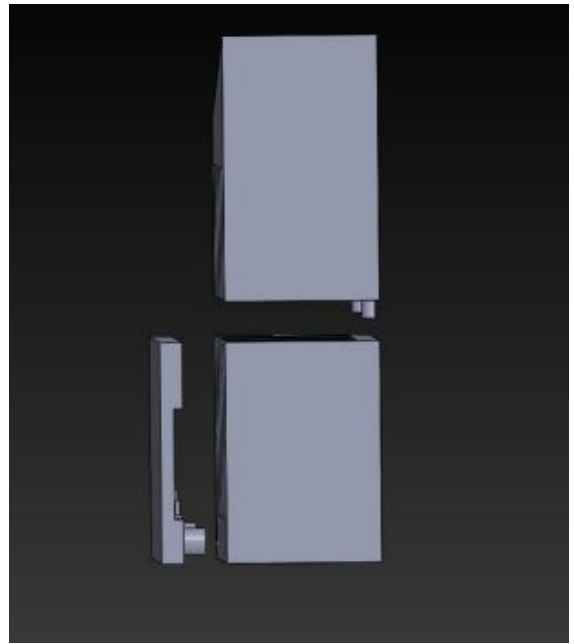
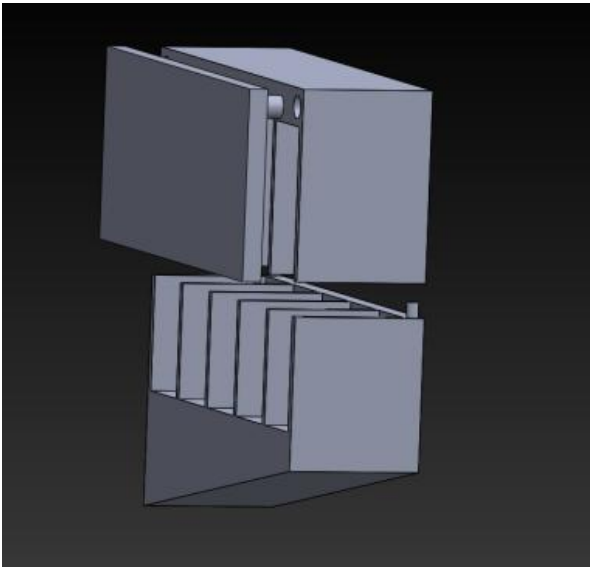
Work Scope



The three key areas under consideration for this project are cost, time, and performance. Cost is the least situational area. It is mainly a 3D printing filament material. The overall price range of the product will likely be 5-\$10. Lots of time will be put into the project. Time will be focused on the CAD design process and calculations. To compensate for this, most of the time will be spent on calculations and errors. With lots of time put into it, it is essential to have high expectations of the final product's performance.

Exploded View of Assembly:





Conclusion

The design validation summary reveals mixed results across three input categories: Portability, Functionality, and Sorting. Portability passed both tests—weight (200g) and size (comparable to a phone)—indicating good compactness and ease of transport. However, Functionality failed both evaluations due to coin jams (working only with Pennies and Dimes) and insufficient bore tolerances in the assembly. In the Sorting category, the main sorting plate failed due to poor 3D print quality and a nozzle issue that hinders coin ejection, while the coin sorting itself passed, although ejection doesn't work for Quarters and Nickels. Overall, only 3 out of 6 tests passed. Ultimately this was a project with not enough time and tools given, with the right materials and finer printing options we could've made a more functional product. There are no citations or references used in this report.